

Daniel Palacios

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 🔄 DanPalBCM

Summary

Ph.D. candidate in Quantitative and Computational Biosciences and NSF Graduate Research Fellow applying AI and machine learning — LLM-powered agentic systems, multi-agent architectures, and EHR foundation models — to clinical informatics and computational biology. Prior research in space radiation biology and NLP at NASA, and in theoretical superconductivity.

Education

PhD	Baylor College of Medicine , Quantitative and Computational Biosciences	Houston, TX
	- NSF Graduate Research Fellow (2024) - NLM Fellow (2023) - Dean's Award of Excellence (2024)	2022 – 2027
BS	University of Houston , Physics and Mathematics (double major)	Houston, TX
	- GPA 3.95/4.00, magna cum laude - Dean's Distinguished Scholar (2018-2022)	2018 – 2022

Experience

Peraton , AI Developer and Research Design Intern	Reston, VA (Remote)
- Cyber Operations, Cyber and Intelligence Sector; mentor Ebenezer Dadson. - Building software powered by large language models and multi-agent systems. - Designing evaluation protocols and controlled experiments measuring how design parameters and system factors affect performance and reliability.	Aug 2026 – present 2 months
Baylor College of Medicine , Graduate Student Researcher	Houston, TX
- Liu Lab, Data Science Center, Duncan Neurological Research Institute, Texas Children's Hospital; advisor Dr. Zhandong Liu. - ClinPreAI: agentic AI system (research, planning, coding, debugging, and interpretation modules) for early postpartum depression risk prediction (EPDS ≥ 10, 31% prevalence) from a single antepartum hospitalization; 4,161 patients, 27 structured variables plus social worker notes; F1 0.68 structured / 0.65 multimodal, matching a custom LLM-XGBoost pipeline. - Pediatric diabetes (T1D/T2D): PySpark pipeline on Palantir Foundry built from OMOP CDM and Epic Clarity EHR data; T2D $n=2,489$, T1D $n=4,018$; 7 binary outcomes at 2- and 5-year horizons; multi-agent LLM clinical-reasoning architectures compared against 8 classical machine-learning and deep-learning baselines. - Trofinetide / Rett syndrome: dual-LLM consensus pipeline extracting 18 treatment-response phenotypes from 26,763 notes (108 patients); 84.2% accuracy and F1 0.674 on the expert-annotated development set. - Clinical-note de-identification: benchmarked 8 LLMs against Stanford TiDE, OpenMed PII, and pattern baselines on 100 expert-annotated pediatric oncology notes (5,322 PHI spans); best single-pass F1 0.918 $\pm$ 0.001; introduced "institutionally-situated PHI"; used LLM outputs to surface 414 candidate gold-standard annotation gaps. - TEDDY: contributed to a 1.84M-parameter generative pediatric EHR foundation model trained on approximately 1.6M children and 73M ICD-10 codes; median first-occurrence AUC 0.72 across 797 disease-onset tasks.	2022 – present 4 years

NASA Ames Research Center, Research AssociateMountain View, CA
May 2022 – Aug 2022
4 months

- Space Biosciences Division; mentor Dr. Jessica Lee. Collaboration continued remotely through August 2026.
- Developed the AMMPER model simulating cell damage under space radiation, validated against BioSentinel mission data.
- Added gamma radiation and alamarBlue metabolic-dye dynamics (Michaelis-Menten kinetics) and a 3D free-radical diffusion model.
- Mentored two undergraduates who carried the project forward; 4 presentations.

NASA Johnson Space Center, Machine Learning Researcher InternHouston, TX
2021 – 2022
1 year

- Mentor Terry Hill.
- Built an NLP pipeline over historical engineering failure reports and an LDA-BERT taxonomy for space-hardware failure reporting.
- Delivered Power BI dashboards and interactive knowledge graphs (INDRA-EIDOS, spaCy, Bokeh-networkx).
- Identified 5-15 critical engineering processes; 8 presentations.

University of Houston, Undergraduate ResearcherHouston, TX
2020 – 2021
1 year

- Advisor Dr. Pavan Hosur.
- Theoretical work on asymmetric superconductivity under broken symmetries; modeled asymmetric supercurrents in non-centrosymmetric superconductors.
- 1 conference presentation; 1 Physical Review B publication.

Publications

Proximity-induced equilibrium supercurrent and perfect superconducting diode effect due to band asymmetry Pavan Hosur, <i>Daniel Palacios</i> 10.1103/PhysRevB.108.094513 (Physical Review B)	2023
Taxonomical modeling and classification in space hardware failure reporting <i>Daniel Palacios</i> , T. R. Hill 10.1038/s41598-026-36813-7 (Scientific Reports)	2026
Factors associated with postpartum depression symptoms following antepartum hospitalization A. N. Goulding, <i>Daniel Palacios</i> , et al. 10.1002/pmf2.70319 (Pregnancy, 2, e70319)	2026
MedSDoH: A Rule-Based System for Social Determinants of Health Extraction from Multi-site EHRs based on the OHNLP Framework J. Ahn, S. Fu, <i>Daniel Palacios</i> , et al. 10.64898/2026.04.27.26351699 (Journal of the American Medical Informatics Association (under review))	2026
ClinPreAI: An Agentic AI System for Early Postpartum Depression Risk Prediction from Multimodal EHR Data <i>Daniel Palacios</i> , et al. 10.1101/2025.11.14.25340265 (PLOS Digital Health (under review))	2026

Institution-Specific LLM Prompting Recovers PHI That De-identification Systems and Their Gold Standards Both Miss 2026
Daniel Palacios, et al.
[10.48550/arXiv.2608.17051](https://doi.org/10.48550/arXiv.2608.17051) (Journal of the American Medical Informatics Association (under review))

Documenting Trofinetide Treatment Phenotypes in Rett Syndrome with a Dual-LLM Consensus Pipeline
Daniel Palacios, et al.
 (Journal of the American Medical Informatics Association (under review; no public preprint))

TEDDY: A Pediatric Foundation Model for Risk Forewarning from ICD-Coded Diagnostic Histories 2026
 M. B. Neeley, J. Botas, J. Jia, L. Yao, *Daniel Palacios*, et al.
[10.48550/arXiv.2607.14191](https://doi.org/10.48550/arXiv.2607.14191) (arXiv:2607.14191 (preprint))

Research Software

Original, public: clinical-clinpreai-benchmarks; clinical-antepartum-ppd-prediction; clinical-trofinetide; clinical-diabetes-t1d-t2d; ml-nasa-nlp; superconducting-diode-effect

Forks: AMMPER; MedSDoH; OverlapPlots (working copies; upstream licenses)

Private: tch-mentormatching (private fork, available on request); LiuzLab/phi-scrubber-evaluation (cite-only, resolves with access)

Teaching and Leadership

Teaching Assistant, Baylor College of Medicine 2023 – 2025

- QCMB 1 (2023-2024); QCMB 2 (2025).
- GSBS Outstanding TA in Quantitative and Computational Biosciences (2025).

Graduate Student Council Representative, QCB Program 2024 – present

- Chair, International Student Committee.
- Career Development Committee; Newsletter.

Research mentor 2022 – 2025

- 7 trainees: 2 NASA Ames interns, 1 PREP, 1 SMS, 3 ASPIRE.

Service

- Keck Seminar Committee (2024)
- Lathisms Committee (2024)
- Scribe, Lunar Biology Technology Workshop, NASA (2022)
- SACNAS Officer, University of Houston (2021)

Grants

BCM Seed Grant: PI: Alison Goulding — awarded

dkNET NIDDK Grant: PI: Dr. Tosur — awarded

Awards and Honors

2025: GSBS Outstanding TA in Quantitative and Computational Biosciences, Baylor College of Medicine
2024: Dean's Award of Excellence, Baylor College of Medicine
2024: NSF Graduate Research Fellowship
2023: NLM Fellowship T15LM007093-32
2023: Lathisms Scholarship
2023: ASM Future Leaders Mentoring Fellowship
2022: ADDSTEAM Scholarship
2022: IMSD NIH-NIGMS Grant 5R25GM056929-25
2022: Magna Cum Laude, University of Houston
2021: Charles P. Benner Scholarship
2021: Provost's Undergraduate Research Scholarship, University of Houston
2021: British American Foundation of Texas Scholarship
2020: Summer Undergraduate Research Fellowship, University of Houston
2018-2022: Dean's Distinguished Scholar, University of Houston
2018-2022: Academic Excellence Scholarship, University of Houston

Selected Presentations

- Goulding, A. N., **Daniel Palacios**, et al. "Factors associated with postpartum depression symptoms following antepartum hospitalization." Texas Children's Research Symposium. Poster. 2026.
- **Daniel Palacios**, et al. "AI-Assisted Construction of Comprehensive Pediatric Type 1 and Type 2 Diabetes Cohorts..." TMC AI Summit. Poster. 2026.
- **Daniel Palacios**, et al. "ClinPreAI: An Agentic AI System for Early Postpartum Depression Risk Prediction..." TMC AI Summit. Poster. 2026.
- Invited talks: NASA Ames Space Biosciences Research Branch (Aug 2022); NASA Johnson Space Center Engineering Directorate (May 2022); NASA Langley ML/AI Community Group (Jul 2021).

Certifications and Training

Graduate Tools and Models in Data Science: Rice University
Machine Learning Certificate: Hewlett Packard Enterprise Data Science Institute
Big Data Technology Fundamentals: Amazon Web Services / NASA
Data Analysis Concepts: Percipio / NASA
ETS Major Field Test in Mathematics: 198/200

Skills

Languages: Python (6+ yrs), R (3+), SQL (3+), MATLAB (2+), Java (1), C++ (1)
Libraries and frameworks: NumPy, pandas, SciPy, scikit-learn, PyTorch, Matplotlib, Seaborn, PySpark, LangChain, LangGraph, Optuna, Hugging Face Transformers, spaCy, NLTK
Machine learning: Deep learning (CNNs, LSTMs, Transformers), PEFT/QLoRA, prompt engineering, retrieval-augmented generation, agentic AI and multi-agent design, AutoML
Statistics and methods: Experimental design, hypothesis testing, feature engineering, statistical modeling
Mathematics: Probability, linear algebra, ODEs/PDEs, optimization
Platforms and tools: AWS (SageMaker, Bedrock), Palantir Foundry, Docker, Git, Power BI